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Scan → 3D in minutes

We kept asking ourselves a simple question: **why does bringing AR to children's books still feel like a studio production?** Unity projects, scene setup, 3D tools, code... it's a lot. What if you could just point your phone at a picture, take one photo, and—after a short wait—watch that character pop up on the *same* page in AR?

So we drew a line in the sand: **no Unity know-how for end users, no modeling tools, no code.** Just *point → click → wait a bit → play*.

What we want people to get out of it

- **Exploration.** Characters “rise” from the page so you can walk around them.
- **Creativity without overhead.** Authors and illustrators add a new layer to their worlds without hiring a 3D team.
- **Scale.** Any flat, clear 2D picture can be a trigger.

The first ship

1. Tap **Generate**.
2. The app snaps a photo of the book page.
3. We send that photo to an image-to-3D service (Meshy) and poll for progress.
4. When the GLB is ready, we load it and anchor it on the **same** 2D page with AR image tracking.

The gotcha we embraced early: we track 2D images, not 3D objects. After the model is generated, the camera must see the same printed picture again (or the same image on a screen/print) before the model snaps into place.

How the pieces talk (first iteration)

- **GenerateController** starts the flow, shows progress, and keeps the screen awake.
- **CameraCapture** takes one good still → gives us JPG bytes (for upload) and a `Texture2D` (for tracking).
- **RuntimeImageLibraryController** adds that texture into a **mutable** runtime image library and remembers the current reference name.
- **MeshyClient** creates the job and polls progress from 0 → 100%.
- **GltfLoader** downloads + loads the GLB into an *inactive* `GameObject` (we keep it hidden until tracking locks).

09/10/2025

- **ModelOverlayController** listens for `ARTrackedImageManager.trackedImagesChanged`. When it sees our reference **and** the state is `Tracking`, it parents the model to the tracked image, scales to the page width, and only then makes it visible.
- **UIController** keeps users in the loop: loading, cancel, retry, and a clear "Model ready ✓".

Early bumps (and what fixed them)

- **"Model ready, but nothing appears."** Usually the camera wasn't seeing the original 2D picture again → no tracking event → no attach. We added a friendly hint: *"Point at the same image to place the model."*
- **Weak reference images.** Low contrast, glare, blur, or repeating patterns score poorly. Our rule of thumb: sharp edges, decent contrast, and a physical width around **10–15+ cm** help a lot.
- **Runtime add needed a nudge** on some devices. We built a small utility to flip `ARTrackedImageManager` off/on or reassign the runtime library after `ScheduleAddImageJob`.